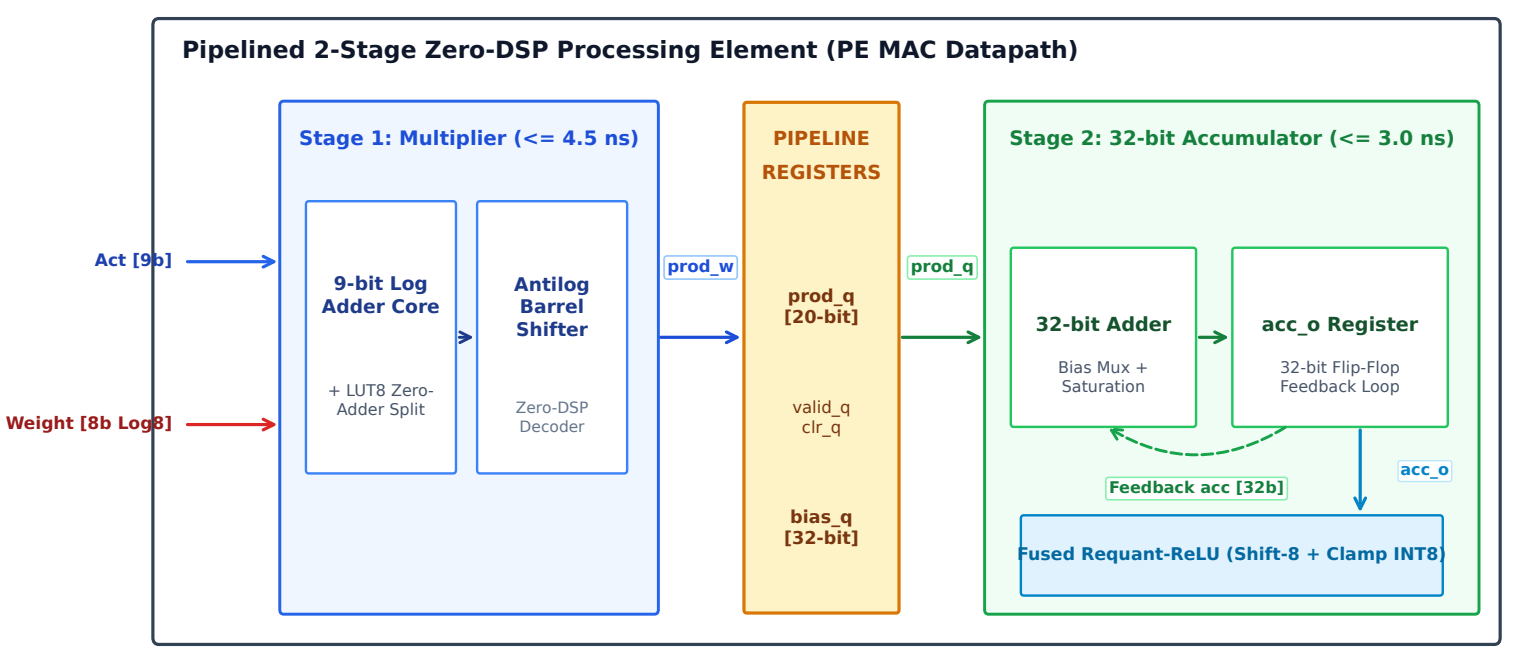
Pipelined 2-Stage Zero-DSP Processing Element (PE MAC Datapath)
Act [9b]
Weight [8b Log8]
Stage 1: Multiplier (<= 4.5 ns)
9-bit Log Adder Core
+ LUT8 Zero-Adder Split
Antilog Barrel Shifter
Zero-DSP Decoder
prod_w
PIPELINE REGISTERS
prod_q [20-bit]
valid_q
clr_q
bias_q [32-bit]
prod_q
Stage 2: 32-bit Accumulator (<= 3.0 ns)
32-bit Adder
Bias Mux + Saturation
acc_o Register
32-bit Flip-Flop Feedback Loop
Feedback acc [32b]
acc_o
Fused Requant-ReLU (Shift-8 + Clamp INT8)